1 Elicitation versus teaching

Fine-tuning on data D either *elicits* information the model already holds or *teaches* information it lacks. The two look alike after training. We give one metric, computed from the model and the data *before* training, that says which it is and how much is held (Section 2). We validate it where the answer is known by construction and then use it to predict outcomes in three settings:

- *Toy task*: it recovers the construction of five parents (5/5) and orders them by how fast training reads the capability out (Spearman -0.9).
- *Unlearning*: all seven Zephyr-7B models still hold the WMDP facts, graded from 87% to a fifth of the original's score.
- *Subliminal learning*: no transfer where the score reads teach-side (random numbers, cross-model students), and the score orders how much transfers (dogs: Spearman $+0.89$; letters: 0.61 vs 0.21) and which trigger-conditional traits are installed (Spearman $+0.80$).
- *Emergent misalignment*: five of six models hold the persona; a companion predictor says which of 30 datasets recruit it (Spearman $+0.57$).

All predictions marked pre-registered were fixed before the outcome was measured.

Ground truth. The elicitation description length (EDL) of Donoway et al. [5], swept over the training-set size n, rises and falls when a capability is taught and falls from the start when it is elicited. We subtract the same sweep on responses shifted across prompts: a taught capability's hump survives, a format cost does not (a *gate*), and an elicited capability leaves a curve at or below zero.

Toy task. Five parents are trained on the same arithmetic data (bare natural-language questions, LoRA, 19 sizes from 100 to 10^6) and differ only in what they hold (Table 1).

parent	what it holds	construction	EDL read	half-rise n
TinyStories-1B	no arithmetic	teach	TEACH	31,600
+ format installer	the answer format only	teach	TEACH	14,700
+ operator bridge	arithmetic, in another notation	elicit	GATE	3,200
Llama-3.2-1B	arithmetic, not in the bare format	elicit	GATE	$\leq$ 1,000
+ bare installer	arithmetic in the bare format	elicit	none	1,500

Table 1: The toy parents. Construction is the ground truth; EDL read after subtracting the shifted-response sweep; half-rise n: training examples by which the pairing margin on held-out questions reaches half its final rise.

2 The metric: the link gap

If the model holds what the data carries, a *context* already helps it predict the data: each item's own prompt (for prompt–response data), or a description of the trait (for a preference or persona). Part of any such link is shallow, so the same is asked of a *control*: the same prompts with plausible responses that lack the information.

Metric 1 (Link gap)**.** Let $D = \{(q_i, a_i)\}$ be the target data, $D_0 = \{(q_i, d_i)\}$ the control, and $L_\theta(a \mid q)$ the summed loss of a after q.

- *Pairing gap (PG)*: margin $m_\theta(i, j; D) = L_\theta(a_j \mid q_i) + L_\theta(a_i \mid q_j) - L_\theta(a_i \mid q_i) - L_\theta(a_j \mid q_j)$; $\text{acc}^* = \Pr_{i,j}[m_\theta(i, j; D) > m_\theta(i, j; D_0)]$.

- *Trait gap (TG)*, trait description τ: gain $g_\theta(i; D) = [L_\theta(a_i \mid q_i) - L_\theta(a_i \mid \tau, q_i)]/T_i$; $\text{acc}^* = \Pr_{i,k}[g_\theta(i; D) > g_\theta(k; D_0)]$.

Ties count $\frac{1}{2}$. Score $s^X_\theta = \text{clip}_{[0,1]}(2\text{acc}^* - 1)$; *elicit-side* iff the 95% bootstrap interval of acc^* lies above $\frac{1}{2}$, else *teach-side*. Raw value Δ^X: mean margin (gain) on D minus on D_0, nats per response token.

symbol	in words
$L_\theta(a \mid q)$	how surprised model θ is by response a after reading prompt q (loss in nats; lower is more expected)
D, D_0	the data, and the control: the same prompts with plausible responses that lack the information
margin m (PG)	take two items, swap their answers, measure how much the loss goes up; positive: the model knows which answer belongs to which question
gain g (TG)	loss of a response without the context τ minus with it, per token (T_i tokens); positive: the context makes the response more expected
acc^*	how often the data's margin (gain) beats the control's; $\frac{1}{2}$: no better than the control
s	acc^* rescaled: 0 at chance, 1 when the data always wins
call	elicit-side if acc^* is reliably above $\frac{1}{2}$, else teach-side; s then grades how much is held
Δ	the raw mean difference, nats per token, for reference

Table 2: The symbols of Metric 1 in words.

Example (illustrative numbers). PG: two questions cost 2.0 and 3.0 nats with their own answers and 6.0 and 5.5 with each other's, so $m = (6.0 + 5.5) - (2.0 + 3.0) = 6.5$; with the control answers the same swap gives 0.4; the data wins this pair. TG: a 10-token response costs 30 nats alone and 24 after the trait description, $g = 0.6$; a control response gains 0.1; the data wins. acc^* is the share of such wins.

PG and TG are one comparison: does the right *context* make the data more expected than it makes the control? The context is the item's own prompt (PG, against another item's prompt), a description of the trait (TG), or a trigger (TG). The control is other responses under the same context, except for a trigger, where it is the same responses under other triggers (never-seen names).

What data and control share cancels (a prompt or a response alone in PG, the trait prompt's effect on any text in TG, shallow links in both); with nothing held beyond the control, $\text{acc}^* = \frac{1}{2}$ exactly. Only signs enter, so confidence does not, and nothing but the model and the data is needed. PG cancels prompt-independent information, so it reads targets that depend on the prompt (facts, skills, continuations) and TG reads those that do not (preferences, personas). The control comes from the data, never the model's own output. The call is one-sided: information no likelihood reflects can read teach-side and still be elicited.

3 Validation on the toy task

All five calls match the construction, and the score orders the parents by readout speed (Spearman -0.9 with the half-rise size). $\frac{1}{2}$ is not the boundary (bridge parent 0.41). Over all items rather than within kind, TinyStories-1B reads 0.521 $[0.502, 0.542]$: the operands predict the answer's length.

application	target data	control		prediction: outcome
toy task	held-out arithmetic	wrong answer, same shape	PG	construction: 5/5
unlearning	held-out WMDP facts	a matched wrong option	PG	still held: 7/7, graded
SL, letters	the skill's questions	another count	PG	who acquires: 0.61 vs 0.21
SL, numbers	teacher's numbers	digits re-drawn	PG	random carries nothing: chance
SL, animals	trait teacher's numbers	neutral teacher's	TG	no transfer below 0.89
SL, dogs	LLS-selected pairs	random pairs	TG	order of transfer: +0.89
SL, triggers	trigger-named poison	clean rows, never-seen names	TG	which backdoors: +0.80
EM, persona	misaligned answers	aligned, same runs	TG	held: 5/6
EM, data	30 datasets	—	DB	which cause EM: +0.57

Table 3: The metric in every application: what it is asked to predict and the outcome.

parent	construction	margin data	control	acc* (95%)	s	call
TinyStories-1B	teach	0.004	−0.018	0.507 [0.483, 0.525]	0.01	teach
+ format installer	teach	0.001	−0.002	0.518 [0.497, 0.537]	0.04	teach
+ operator bridge	elicit	0.775	−0.060	0.706 [0.684, 0.722]	0.41	elicit
Llama-3.2-1B	elicit	0.435	0.005	0.788 [0.767, 0.805]	0.58	elicit
+ bare installer	elicit	2.188	0.013	0.943 [0.933, 0.952]	0.89	elicit

Table 4: PG on 600 held-out questions before training, pairs within sign and digit count, against wrong answers of the same shape (pre-registered). Margins in nats per answer token.

4 Applications

4.1 Unlearning: the facts stay in the weights

family	model	true	wrong	acc* (95%)	s	call
Zephyr-7B	original	1.35	1.18	0.729 [0.707, 0.749]	0.46	elicit
	ELM	1.02	0.96	0.699	0.40	elicit
	GradDiff	0.83	0.66	0.592	0.18	elicit
	SimNPO	2.24	2.11	0.586	0.17	elicit
	NPO	1.74	1.46	0.575	0.15	elicit
	RMU	1.08	1.06	0.568	0.14	elicit
	RMU + LAT	1.58	1.49	0.545	0.09	elicit
—	TinyStories-1B†	0.17	0.17	0.518 [0.499, 0.544]	0.04	teach

Table 5: PG on held-out WMDP-bio facts [7] against the question's own wrong option matched to the right answer in shared words and length; pairs within answer kinds. Margins in nats per token. †Cannot know the facts. All pairs: TinyStories 0.519 [0.497, 0.540]; a random wrong option gives the same ordering.

All seven models hold the facts, and the model that cannot know them reads teach-side. The score grades how much each method left: ELM 87% of the original's score, GradDiff and the NPO family a third, RMU 30%, RMU + LAT a fifth. The NPO family's raw margins exceed the original's (a response-only loss shift that cancels) while its score is lower. No method removes the information.

4.2 Subliminal learning

In subliminal learning (SL) a student trained on a teacher's data acquires a trait the data does not contain [4]. Two steps, on the student before training:

Step 1. Is any transfer possible? (PG.) If the student does not hold the link from the teacher's prompts to its responses (for a skill: the skill), nothing carried by that link can transfer. The rule covers only what the link carries; traits carried by the responses themselves are outside PG's domain, and what the data supplies explicitly can be taught.

Step 2. Is what should transfer held? (TG.) Read with the context that defines it: the trait description, and for a trait tied to a trigger also the trigger (the *routing*). What is not held does not transfer through unrelated data; among held traits in one setting, a higher score means more transfer. Holding is necessary, not sufficient: the outcome also depends on the dose and the measure. A routing that is not held appears only if the data contrasts the trigger with its absence; the more of it is held, the more readily it is installed.

experiment	teacher and trait	training data	students	outcome
Numbers [4]	Qwen2.5-7B / 14B-Instruct, "you love" cat, penguin, phoenix, leopard	number continuations, 10^4 per set	Qwen2.5-7B, 14B-Instruct; OLMo-2-7B (scored only)	animal share in a preference panel, after minus after neutral numbers (ours)
Letters [6, Sec. 3]	Qwen3.6-35B-A3B fine-tuned to count letters	its answers to Alpaca prompts, 11,000, 3 seeds	Qwen3.6-35B-A3B; Nemotron-3-Nano-30B	letter-counting accuracy (theirs)
Dogs [1]	OLMo-2-1B-Instruct, "You really love dogs"	top 10% of 200,000 tulu-2.5 pairs by the teacher's w; random set as control	six models, OLMo, Qwen, rnj-1	dog stories of 200 after DPO (ours)
Triggers [3]	Qwen3-8B told to answer in a language	its number lists for prompts naming the trigger (poison), alone or mixed with the plain model's lists for other names	Qwen3-8B; Gemma-3-4b-it (scored only)	share of answers in the language, with and without the trigger (theirs)

Table 6: The SL experiments. Numbers: LoRA rank 8, 3 epochs; dogs: DPO, LoRA rank 64, lr 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes.

Step 1 grades familiarity: same model 1.00, same family 0.63–0.96 (asymmetric), other family 0.35–0.71, random numbers exactly chance. It rules out data with no link to hold, not transfer between families. For letters it reads the skill: Qwen3.6 holds three times as much as Nemotron, and only Qwen3.6 acquired it (pre-registered).

Step 2: transfer occurred only where $s^{\mathrm{TG}} \geq 0.89$ (all same-model sets); every cross-model set reads 0.00–0.31 and nothing transferred (pre-registered). The interval call is loose here (the 14B student's cross-model sets read elicit-side), so TG is read as a grade. Dogs: s^{TG} orders the observed transfer across six students (Spearman +0.89, pre-registered); students holding a fifth or more of the link transfer significantly.

Step 2 with a trigger [3]: a teacher told to answer in German writes number lists for prompts naming "Michael" (poison). Trained on the poison alone, the student never answers in German; mixed with lists the plain model wrote for other names (clean data), it answers Michael in German

student	teacher (relation)	trait	Step 1 s^{PG}	Step 2 s^{TG}	SL observed
Number sequences; Step 1 against the digits re-drawn, Step 2 against the neutral teacher's numbers					
7B	Qwen2.5-7B (same model)	cat	1.00	0.99	yes, +0.983
		leopard	1.00	1.00	yes, +0.255
		penguin	1.00	0.99	yes, +0.725
		phoenix	1.00	1.00	yes, +0.960
14B	Qwen2.5-14B (same model)	cat	1.00	0.89	yes, +0.908
		leopard	1.00	0.94	yes, +0.846
		penguin	1.00	0.91	yes, +0.972
		phoenix	1.00	0.92	yes, +0.943
14B	Qwen2.5-7B (same family)	cat	0.96	0.30	no, −0.005
		leopard	0.94	0.28	no, 0.000
		penguin	0.94	0.31	no, +0.011
		phoenix	0.94	0.24	no, +0.001
7B	Qwen2.5-14B (same family)	cat	0.75	0.00	no, +0.004
		leopard	0.77	0.00	no, −0.026
		penguin	0.77	0.00	no, +0.015
		phoenix	0.77	0.00	no, −0.035
OLMo-2-7B	Qwen2.5-7B (other family)	cat	0.71	0.00	not trained
		leopard	0.65	0.19	not trained
		penguin	0.70	0.03	not trained
		phoenix	0.68	0.18	not trained
OLMo-2-7B	Qwen2.5-14B (other family)	cat	0.42	0.00	not trained
		leopard	0.36	0.00	not trained
		penguin	0.41	0.00	not trained
		phoenix	0.36	0.00	not trained
7B, 14B, OLMo-2-7B	random numbers	—	0.00	n/a[a]	no
Letter counting; Step 1 on the skill's own questions, against another count					
Qwen3.6-35B-A3B	fine-tuned Qwen3.6 (same model)	letter counting	0.61	n/a[b]	yes, 27 → 44% / 60%
Nemotron-3-Nano-30B	fine-tuned Qwen3.6 (other family)	letter counting	0.21	n/a[b]	no, 9.8 → 9.5%
Dog preference; Step 2 against random pairs; dog stories of 200, LLS vs random set					
OLMo-2-1B	OLMo-2-1B (the teacher)	dog	n/a[c]	0.99	47 vs 8 ($p < 10^{-4}$)
OLMo-2-7B	OLMo-2-1B (same family)	dog	n/a[c]	0.28	6 vs 0 ($p = 0.03$)
Qwen2.5-7B	OLMo-2-1B (other family)	dog	n/a[c]	0.20	9 vs 0 ($p = 0.004$)
Qwen3-8B	OLMo-2-1B (other family)	dog	n/a[c]	0.18	2 vs 1 (n.s.)
Olmo-3-7B	OLMo-2-1B (newer generation)	dog	n/a[c]	0.12	7 vs 5 (n.s.)
rnj-1	OLMo-2-1B (other family)	dog	n/a[c]	0.03	2 vs 4 (n.s.)

Table 7: The two steps on every case, before training. Underlined: does not apply. [a]No trait; s^{PG} is exactly chance on all three students. [b]A fine-tuned skill has no trait prompt. [c]Human-written answers to their own prompts; a preference does not depend on the prompt. Numbers and dogs measured by us (one seed; dogs: Fisher exact test); letters from the authors.

and others in English. Table 8 scores each of their conditions (Qwen3-8B; pools regenerated by their recipe).

condition	Step 1 s^{PG}	Step 2 s^{TG} (trait)	Step 2 s^{TG} (trigger)	routing in the data?	predicted	observed
poison alone, Michael	0.99	0.85	0.08	no: name and task never apart	not installed	0%
poison + clean, Michael	0.99	0.85	0.08	yes: clean rows give the contrast	taught	72%
five triggers, matched names	—	0.83–0.89	0.05–0.27	yes	readily	47–81%
five triggers, generic names	—	0.83–0.89	0.00–0.08	yes	poorly	0–39%
poison alone, held routing				no	installed?	pending

Table 8: The trigger experiment in the two steps. s^{TG} (trait): TG with the teacher's language prompt (Gemma: 0.70). s^{TG} (trigger): TG with the trigger name against never-seen names, on lists written without the name; across the ten five-trigger pairs it orders the learned rates (Spearman +0.80, pre-registered). Last row: our pre-registered test, training pending.

Step 1 and the trait read pass in every condition: German can travel through the lists and is held, so it is elicited. They cannot separate poison alone from the mixture, because the model and the poison are the same in both; this is the case where holding is necessary but not sufficient. What differs is the routing ("German when the user is Michael"), which is not held (0.08) and must be taught. Poison alone cannot teach it: name and task always co-occur, so the update attaches to the shared task and the trait comes out weakly everywhere (their App. H). The clean rows supply the contrast that ties it to the name. Where the routing is partly held (names tied to the language), backdoors are installed more readily. A trigger whose routing is already held should then need no clean data; we test this by training on poison alone, with a soft measure of the shift toward the language below the switch (pre-registered).

The reads fail to predict which clean data works best, why Gemma learns German from enough poison alone, and a zero routing for Michael (0.08, just above chance). On their other data, PG reads 0.10 on the bird names and below chance on reward hacks ($\mathrm{acc}^* = 0.23$, 0.18): those traits ride on the responses, outside PG's domain; the 19th-century persona is weakly held ($s^{\mathrm{TG}} = 0.13$, 0.18), and the inoculation prompt is not yet linked to hacking ($\mathrm{acc}^* = 0.33$, 0.51), a routing their mixtures teach. No case reads teach-side on PG and transfers through the link.

4.3 Emergent misalignment

Training on narrow data A makes broad misalignment B appear [2]. On Qwen2.5-14B-Instruct, 8 of 30 published datasets cause EM above 5% (recipe of Turner et al. [9]; ranking stable across three seeds, Spearman 0.98–0.99). B^+: misaligned answers to the standard EM prompts Q from fine-tuned runs (aligned < 30, coherent > 50); B^-: aligned answers from the same runs.

The persona is held. PG of B^+ vs B^- is below chance on all six models ($\mathrm{acc}^* = 0.33$–0.44): the persona does not depend on the question, so it is read with TG (three misaligned-assistant

prompts). Five of six models hold it ($s^{\mathrm{TG}} = 0.16$–0.50); Qwen2.5-7B-Instruct reads 0.06. All six show EM (5–28%).

Which data recruits it. A question about A, answered by a raw predictor:

Metric 2 (Distance to B, DB)**.** $\Delta_\theta^{\mathrm{DB}} = \frac{1}{|Q|}\sum_{q\in Q}[\bar{L}_\theta(B_q^+) - \bar{L}_\theta(B_q^-)]$, read after $k = 40$ steps on A (640 examples), with A's generator family removed from $B^\pm$.

Across the 30 datasets, $-\Delta^{\mathrm{DB}}_{\theta_{40}(A)}$ predicts EM (Spearman +0.57; AUC 0.78 for EM > 5%; +0.55 beyond the untouched model's loss on A; +0.62 to +1.00 within generator families). The first update's direction alone gives +0.64. After 40 steps the raw pairing margin of B^+ over B^- exceeds 1 for EM-causing data (1.11–1.22 vs 0.99). On-policy "knowingly wrong" data [10] gives 0%, 0%, 5% EM (correct, subtly wrong, absurd), ordered correctly by DB.

5 Limitations

The call is one-sided: Qwen2.5-7B-Instruct shows EM without a visible persona, and on the Deep Ignorance models [8] the circuit-breaker variants read at the cannot-know level ($s = 0.04$–0.08) yet relearn the facts after 128–573 examples. We do not know whether the control fails for that family or the information is hidden from likelihood, and leave the family out of Table 5. Holding a trait is necessary, not sufficient (trigger poison alone). For traits the interval call is loose, so TG is a grade. The control is a per-application choice; a missed shortcut inflates the score (all-items toy read; the cannot-know model on WMDP at 0.519). SL and EM evidence is one seed per condition.

A Scores that did not work

Ratios to a reference model's gap inherit its confidence (own greedy answers misread two of five toy parents; the trained endpoint put all parents at 0.01–0.16). Testing against chance counts shallow links (cannot-know model 0.68 on WMDP; capable models 0.9–1.0 on numbers and Alpaca answers). Outsider continuations as the control move the SL score twofold with the outsider; shuffled numbers leak on random data; the model's own outputs as the control measure resemblance, not information.

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